Chapter 6, Problem 66P

Problem

The current $i(t)$ through a 20-mH inductor is equal, in magnitude, to the voltage across it for all values of time. If $i(0)$ = 2 A, find $i(t)$.

Step-by-step solution

Step 1 of 1

Given

$L = 20\text{mH}$; $i(0) = 2A$

$i(t) = v(t)$ ----------------(1)

We know $v\left(t\right) = L\frac{\text{di}(t)}{\text{dt}}$ ---------(2)

(1) into (2):

$$i\left(t\right) = L\frac{\text{di}(t)}{\text{dt}}$$

$$\frac{1}{i(t)}\text{di}\left(t\right) = \frac{1}{L}\text{dt}$$

By taking integration on both sides

$$\int_0^t \frac{1}{i(t)}\,\text{d}\,i(t) = \frac{1}{L}\int_0^t 1\,\text{d}\,t$$

$$[\text{lni}(t)]_0^t = \frac{1}{L}[t]_0^t$$

$$\text{lni}\left(t\right) - \text{lni}\left(0\right) = \frac{t}{L}$$

$$\text{lni}\left(t\right) - \text{lni}\left(0\right) = \ln\left[e^{\frac{t}{L}}\right]$$

$$\text{lni}\left(t\right) - \ln\left(2\right) = \ln\left[e^{50t}\right]$$

$$\text{lni}\left(t\right) = \ln\left[2e^{50t}\right]$$

$$i\left(t\right) = 2e^{50t}A$$